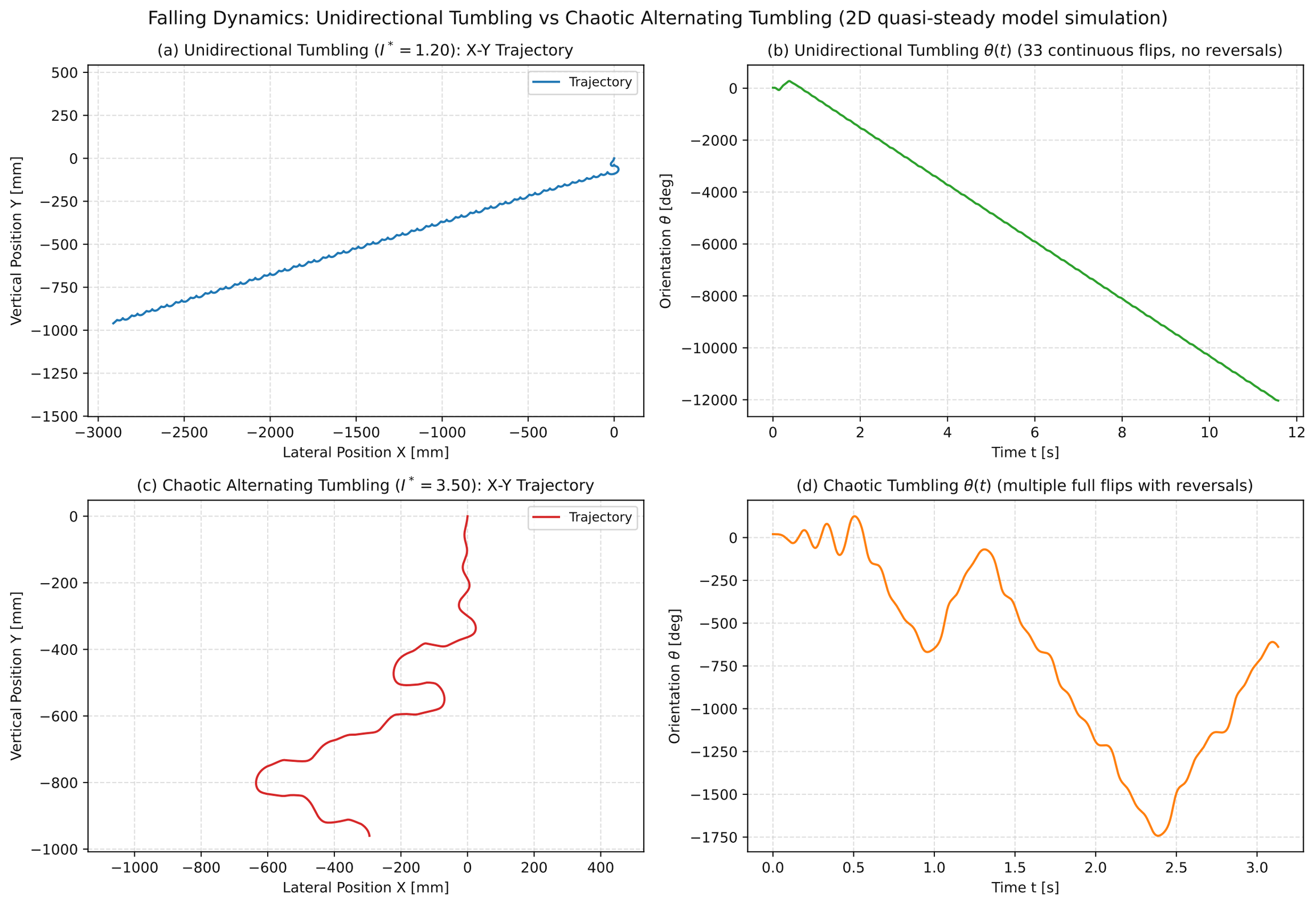
Falling Dynamics: Unidirectional Tumbling vs Chaotic Alternating Tumbling (2D quasi-steady model simulation)
(a) Unidirectional Tumbling ($I^* = 1.20$): X-Y Trajectory
Trajectory
Vertical Position Y [mm]
Lateral Position X [mm]
(b) Unidirectional Tumbling $\theta(t)$ (33 continuous flips, no reversals)
Orientation θ [deg]
Time t [s]
(c) Chaotic Alternating Tumbling ($I^* = 3.50$): X-Y Trajectory
Trajectory
Vertical Position Y [mm]
Lateral Position X [mm]
(d) Chaotic Tumbling $\theta(t)$ (multiple full flips with reversals)
Orientation θ [deg]
Time t [s]